

Ratio, Proportion and Rates of Change

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the ratio 12 : 18 in its simplest form.

.....

2 [2 marks]

Share £60 in the ratio 2 : 3

.....

3 [2 marks]

A recipe for 4 people uses 200 g of rice.
How much rice is needed for 10 people?

.....

4 [2 marks]

Work out 15% of 240

.....

5 [3 marks]

A map is drawn to a scale of 1 : 25 000
Two villages are 6 cm apart on the map.
Work out the real distance between the villages, in kilometres.

.....

6

[3 marks]

- (a) Increase £48 by 25%
- (b) Decrease 90 kg by 30%
-

7

[3 marks]

A 500 g box of cereal costs £1.80

An 800 g box of the same cereal costs £2.72

Which box is better value for money? You must show your working.

8

[3 marks]

A car travels 150 km in 2 hours 30 minutes.

Work out the average speed of the car in km/h.

9

[4 marks]

In a class the ratio of boys to girls is 3 : 5

There are 12 more girls than boys.

Work out the total number of students in the class.

10

[4 marks]

Here is a recipe for 12 biscuits.

Flour	Butter	Sugar
150 g	80 g	60 g

Kim has 400 g of flour, 200 g of butter and 200 g of sugar.

What is the greatest number of biscuits Kim can make?

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Write the ratio 4 : 10 in the form 1 : n

2

[2 marks]

Express 45 minutes as a percentage of 3 hours.

3

[3 marks]

In a sale, the price of a jacket is increased by 20%.

The new price is £54

Work out the original price of the jacket.

4

[3 marks]

Ravi invests £2000 in an account paying 3% compound interest per year.

Work out the value of the investment after 4 years.

Give your answer to the nearest penny.

5

[4 marks]

y is inversely proportional to x^2 .

When $x = 2$, $y = 9$

(a) Find a formula for y in terms of x .

(b) Work out y when $x = 6$

(c) Work out the positive value of x when $y = 4$

6

[4 marks]

A bag contains red counters and blue counters in the ratio 5 : 3
10 more blue counters are put into the bag.
The ratio of red to blue counters is now 5 : 4
Work out the total number of counters now in the bag.

7

[4 marks]

Two solids are mathematically similar.
The smaller solid has surface area 50 cm^2 and volume 40 cm^3 .
The larger solid has surface area 200 cm^2 .
Work out the volume of the larger solid.

8

[4 marks]

A block of metal has mass 2.4 kg and volume 300 cm^3 .
(a) Work out the density of the metal in g/cm^3 .
(b) A second block of the same metal has volume 750 cm^3 . Work out its mass in kilograms.

9

[4 marks]

A lorry drives 120 km at an average speed of 80 km/h.
It then drives a further 60 km at an average speed of 40 km/h.
Work out the average speed for the whole journey.

The population of a town is 40 000.

The population is falling by 4% each year.

- (a) Work out the population after 5 years, to the nearest whole number.
- (b) Find the least number of complete years it takes for the population to fall below 30 000.

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